# Notes of 'On a problem of Brown, Erdős and Sós'

Sifei Li

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## 1 Proof of Upper Bound

Notation: in an $r$-graph $\mathcal{F}$, we use $\mathcal{F}$ to represent the edge set $E(\mathcal{F})$, $|\mathcal{F}|$ to represents the number of edges in $\mathcal{F}$, and $e \in \mathcal{F}$ to represent that $e$ is an edge of $\mathcal{F}$. Also, for a hyperedge $e$, we use $V(e)$ to represent its vertices. We also follow the terminologies from [LS23]: A $k$-configuration denotes a $(k(r-t)+t,k)$-configuration, a $k^-$-configuration denotes a $(k(r-t)+t-1,k)$-configuration. A hypergraph is $k$-free (resp. $k^-$-free) if it does not contain any $k$-configuration (resp. $k^-$-configuration). The component mentioned in the following proofs refers to a t-tight component defined in the paper [LS23].

**Lemma 1** (Lemma 2.6 in [LS23])**.** *Let $r, k$ and $t$ be fixed positive integers. Let $\mathcal{F}$ be a $k$-free $n$-vertex $r$-graph. Then there exists a subhypergraph $\mathcal{F}'$ of $\mathcal{F}$ such that the following holds.*

*(P1) $\mathcal{F}'$ is $\ell^-$-free for every $\ell \in [2,k]$ with $\ell \mid (k-1)$ or $\ell \mid k$;*

*(P2) there is no $3^-$-configuration in $\mathcal{F}'$ which contains a 2 -configuration;*

*(P3) for every positive integers $a$ and $b$ with $a+b=k$, every $a^-$-configuration and every $b$-configuration of $\mathcal{F}'$ are edge-disjoint;*

*(P4) $|\mathcal{F}'| \geq |\mathcal{F}| - O\left(n^{t-1}\right)$.*

**Theorem 1.** *Let $k, t, r$ be integers, $k \geq 5$ be odd, $t \geq 2$, $r$ is large enough such that*

$$[2r-t(2k-3)]^t - 2[r-(k-2)t]^t - \frac{3}{2}r^t \geq t! \left[(k-3) + (k-5)(k-2)\binom{4t-4}{t} - \frac{3}{4}\right],$$

*, then*

$$\limsup_{n\to\infty} n^{-t} f^{(r)}(n; k(r-t)+t, k) \leq \frac{1}{t!}\frac{2}{2\binom{r}{t}-1}.$$

*Proof.* Let $F$ be a $k$-free $n$-vertex $r$-graph. By lemma 1, $\exists \mathcal{F}' \subseteq \mathcal{F}$ such that $\mathcal{F}'$ satisfies $(P1), (P2), (P3)$ and $|\mathcal{F}'| \geq |\mathcal{F}| - O\left(n^{t-1}\right)$.

Denote $\mathcal{F}_1 = \{e : e \in \mathcal{F}', |V(e) \cap V(f)| < t, \forall f \in \mathcal{F}', f \neq e\}$ as the set of edges of $\mathcal{F}'$ that belong to components of size 1. Denote $\mathcal{F}_2$ as the set of edges of $F'$ which belong to components of size 2. If $e \in \mathcal{F}_2$, then $\exists f \in \mathcal{F}'$ such that $|V(e) \cap V(f)| \geq t$ and for any other $f' \in \mathcal{F}', |V(e) \cap V(f')| < t$. $\mathcal{F}_3 := \mathcal{F}' \backslash (\mathcal{F}_1 \cup \mathcal{F}_2)$.

Denote $\mathcal{G}_1$ as the $t$-shadow of $\mathcal{F}_1, \mathcal{G}_2$ as the $t$-shadow of $\mathcal{F}_2$, and $\mathcal{G}_3$ as the collection of t-sets $T$ such that $T \notin \mathcal{G}_1 \cup \mathcal{G}_2$, and there is a unique component in $\mathcal{F}_3$ that contains a 2-configuration whose vertex set contains $T$.

Notice that $\mathcal{G}_1, \mathcal{G}_2, \mathcal{G}_3$ are disjoint. Let $\alpha = \frac{1}{2}\left(2\binom{r}{t} - 1\right)$.

By Lemma 2 below, for $i = 1, 2, 3, |G_i| \geq \alpha |F_i|$, then

$$\begin{aligned}
|\mathcal{F}'| &= |\mathcal{F}_1| + |\mathcal{F}_2| + |\mathcal{F}_3| \\
&\leq \frac{1}{\alpha}\left(|\mathcal{G}_1| + |\mathcal{G}_2| + |\mathcal{G}_3|\right) \\
&\leq \frac{1}{\alpha}\binom{n}{t} \\
&= \frac{2\binom{n}{t}}{2\binom{r}{t} - 1} \\
&\leq \frac{2n^t}{t!\left(2\binom{r}{t} - 1\right)}
\end{aligned}$$

Therefore, $\limsup_{n\to\infty} n^{-t}|\mathcal{F}| \leq \limsup_{n\to\infty} n^{-t}\left(|\mathcal{F}'| + O\left(n^{t-1}\right)\right) \leq \frac{2}{t!\left(2\binom{r}{t}-1\right)}$. □

Note that we impose the additional condition $k \geq 5$ to avoid certain counting issues. This is because the following proof involves the term $k - 5$. The case $k = 3$ has been proved in a separate paper. The goal of the rest of this note is to prove Lemma 2:

**Claim 1** (Claim 3.2 in [LS23]). *Let $\mathcal{F}$ be a $k$-free $r$-graph and $e \in \mathcal{F}$. Then the number of 2-configurations of $\mathcal{F}$ containing $e$ is at most $k - 2$.*

**Claim 2.** *Suppose that $r, t, k$ are integers that satisfy $t, k \geq 2$ and $r$ is large enough such that*

$$\left\{[2r - t(2k-3)]^t - 2[r - (k-2)t]^t - \frac{3}{2}r^t\right\} \geq t!\left[(k-3) + (k-5)(k-2)\binom{4t-4}{t} - \frac{3}{4}\right],$$

*Let $q = r - (k-2)t$. Then*

$$\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t} \geq \frac{3}{2} \cdot \frac{1}{2}\left(2\binom{r}{t} - 1\right)$$

*Proof.* Since $\frac{n^t}{t!} \geq \binom{n}{t} \geq \frac{(n-t)^t}{t!}$, we have

$$\begin{aligned}\binom{2q}{t} - 2\binom{q}{t} - \frac{3}{2}\binom{r}{t} &\geq \frac{1}{t!}\left\{[2r - t(2k-3)]^t - 2[r-(k-2)t]^t - \frac{3}{2}r^t\right\} \\ &\geq (k-3) + (k-5)(k-2)\binom{4t-4}{t} - \frac{3}{4}\end{aligned}$$

given the assumption of $r$. □

We remark that the assumption of $r$ is feasible when $r$ is large enough, as when $t \geq 2$, the size of left side is $Cr^t + O(1)$ with some positive constant $C$, and the right side is just a constant with respect to $r$.

**Lemma 2.** *For $i = 1, 2, 3, |\mathcal{G}_i| \geq \alpha\,|\mathcal{F}_i|$.*

*Proof.* For $i = 1$, by definition $|\mathcal{G}_1| = \binom{r}{t}|\mathcal{F}_1| \geq \left(\binom{r}{t} - \frac{1}{2}\right)|\mathcal{F}_1| = \alpha\,|\mathcal{F}_1|$, for any $r \geq t$.

For $i = 2$, for every 2-configuration $(e, f)$ where $e, f \in \mathcal{F}_2, |V(e) \cap V(f)| \geq t$, so $|V(e) \cup V(f)| \leq 2r - t$. However, since $\mathcal{F}'$ is $2^-$-free by $(P1)$ of Lemma 1, $\mathcal{F}_2 \subseteq \mathcal{F}'$ is also $2^-$-free. Then any $2(r-t) + t - 1 = 2r - t - 1$ vertices span at most one edge. So $|V(e) \cup V(f)| > 2r - t - 1$, which forces $|V(e) \cup V(f)| = 2r - t$. Thus, the number of choices of $t$-shadow of a 2-configuration $(e, f)$ is $2\binom{r}{t} - 1$. Multiplying by the number of 2-configurations in $\mathcal{F}_2$ results in

$$|G_2| = \left(2\binom{r}{t} - 1\right)\frac{|\mathcal{F}_2|}{2} = \alpha\,|\mathcal{F}_2|.$$

For $i = 3$, note that by definition $\mathcal{F}_3$ can be partitioned into $t$-tight components, each of size at least 3. Let $C$ be a component of $\mathcal{F}_3$. It suffices to show that $|\mathcal{G}_3(C)| \geq \alpha|C|$, where $\mathcal{G}_3(C)$ denotes the restriction of $\mathcal{G}_3$ to the edges in $C$. If this holds for every component, then summing over all components gives the desired result.

By the definition of a $t$-tight component, there are at least $|C| - 1$ distinct 2-configurations contained in $C$.

We first state a few observations about $C$ that will be used later:

- $C$ is $k$-free and $2^-$-free, and no $3^-$-configuration in $C$ contains a 2-configuration. These properties are inherited from $\mathcal{F}$ or follow from Lemma 1.
- Each 2-configuration in $C$ spans exactly $2r - t$ vertices, by the $2^-$-free property.
- $|C| \leq k - 1$, since otherwise this would violate the $k$-free property.

We partition $\mathcal{G}_3(C)$ into two disjoint collections of $t$-sets, denoted by $\mathcal{A}$ and $\mathcal{B}$. The collection $\mathcal{A} \subseteq \mathcal{G}_3(C)$ consists of all $t$-sets that are fully contained in a single edge of a 2-configuration, while $\mathcal{B} \subseteq \mathcal{G}_3(C)$ consists of all $t$-sets $T$ for which there exists a 2-configuration $(e_1, e_2)$ such that

$$T \cap \big(V(e_1) \setminus V(e_2)\big) \neq \emptyset \quad \text{and} \quad T \cap \big(V(e_2) \setminus V(e_1)\big) \neq \emptyset.$$

We only estimate the size of $\mathcal{B}$ and show that this is enough. For each edge $e$ in $C$, it shares at most $t$ vertices with any other edge in $C$, since otherwise the two edges would form a $2^-$-configuration. Using $|C| \leq k-1$, it follows that there are at least $q := r - t(k-2)$ vertices of $e$ that are not shared with any other edge in $C$. For each edge $e$, we randomly choose a subset with $q$ vertices, $Y_e$, from the set $\{v \in V(e) : v \notin V(e') \forall e' \in C, e' \neq e\}$.

For each 2-configuration $S = (e_1, e_2)$ in $C$, define $\mathcal{B}_S := \{T \subseteq V(S) : |T| = t, T \cap Y_{e_1} \neq \emptyset, T \cap Y_{e_2} \neq \emptyset\}$. We have

$$\mathcal{B}_S = \binom{2q}{t} - 2\binom{q}{t},$$

where the term $2\binom{q}{t}$ subtracts the $t$-sets that are fully contained in a single edge. It is also worth mentioning that for two distinct 2-configurations $S$ and $S'$, $\mathcal{B}_S \cap \mathcal{B}_{S'} = \emptyset$.

We need to remove the $t$-sets that are contained in other components. These will be considered in the following claims:

1. The number of $t$-sets $T$ in $(e_1, e_2)$ that are contained in an edge $e' \notin C$ is at most $(k-3)$.
2. The number of $t$-sets $T$ in $(e_1, e_2)$ that are contained in the vertex set of a 2-configuration $(e', f')$ with $(e', f') \not\subseteq C$, and is not contained in any single edge of $(e', f')$ is at most $(k-2)(k-5)\binom{4t-4}{t}$.

To prove 1, suppose there is an edge $e' \notin C$ such that $e'$ shares at least $t$ vertices with $(e_1, e_2)$. Then it must share exactly $t$ vertices with $(e_1, e_2)$, otherwise they would form a $3^-$-configuration containing a 2-configuration, which is impossible.

For a fixed $(e_1, e_2)$, there are at most $(k-3)$ such edges $e'$, since otherwise these edges together would form a $k$-configuration. Therefore, the number of $t$-sets $T$ in $(e_1, e_2)$ that are contained in some edge $e' \notin C$ is at most $(k-3)\binom{t}{t} = (k-3)$.

To prove 2, we first note that neither $e'$ nor $f'$ can intersect any edge of $C$ in more than $t-1$ vertices, otherwise $e'$ (or $f'$) would lie in $C$, which would imply $(e', f') \subseteq C$. Thus $|V(e_1, e_2) \cap V(e', f')| \leq 4(t-1)$.

We now bound the number of such 2-configurations $(e', f')$. Let $R$ be a 3-configuration in $C$ that contains $(e_1, e_2)$, which exists since $|C| \geq 3$.

Let $\mathcal{N}(R,C)$ be the collection of 2-configurations that are not contained in $C$ and whose vertex sets intersect $V(e_1,e_2)$ in at least $t$ vertices. Let $\mathcal{M}$ be a maximal pairwise edge-disjoint subcollection of $\mathcal{N}(R,C)$. We claim that $|\mathcal{M}| \leq \frac{k-5}{2}$.

If $|\mathcal{M}| > \frac{k-5}{2}$, choose $\frac{k-3}{2}$ members $S_1,\ldots,S_{(k-3)/2}$ of $\mathcal{M}$. Then the union $R \cup S_1 \cup \cdots \cup S_{(k-3)/2}$ contains $3 + 2\cdot\frac{k-3}{2} = k$ edges, and at most $(3r-2t)+(2r-t)\frac{k-3}{2} - \frac{k-3}{2}t = k(r-t)+t$ vertices, which forms a $k$-configuration, contradicting the $k$-free assumption. Hence $|\mathcal{M}| \leq \frac{k-5}{2}$.

By maximality of $\mathcal{M}$, every 2-configuration $S \in \mathcal{N}(R,C)$ shares an edge with some member of $\mathcal{M}$. By Claim 1, the maximum number of 2-configurations containing a fixed edge is $k-2$. Thus $|\mathcal{N}(R,C)| \leq 2(k-2)\,|\mathcal{M}| \leq (k-2)(k-5)$.

Since each 2-configuration contributes at most $\binom{4t-4}{t}$ $t$-sets, we obtain the bound in 2.

Therefore, each 2-configuration in $C$ contributes at least

$$\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t}$$

distinct $t$-sets to $\mathcal{B}$.

Multiplying by the total number of 2-configurations in $C$, we have

$$\begin{aligned}
|\mathcal{G}_3(C)| &\geq (|C|-1)\left[\binom{2q}{t} - 2\binom{q}{t} - (k-3) - (k-5)(k-2)\binom{4t-4}{t}\right] \\
&\geq (|C|-1)\frac{3}{2}\alpha && \text{by claim 2} \\
&= |C|\alpha + \frac{1}{2}\alpha|C| - \frac{3}{2}\alpha \\
&\geq |C|\alpha && \text{as } |C| \geq 3
\end{aligned}$$

as desired.

□

## 2 Reference

## References

[LS23] Shoham Letzter and Amedeo Sgueglia, *On a problem of brown, erdős and sós.*